

ARPAN GANGULY

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PROFESSIONAL SUMMARY

Senior Java Backend Engineer / Technical Lead with 9.5+ years of experience building scalable, high-performance distributed systems using Java, Spring Boot, and microservices architecture. Strong expertise in designing cloud-native solutions on AWS and implementing event-driven systems using Kafka. Recently focused on integrating Generative AI solutions, including RAG pipelines, embeddings, and LLM-based services, to deliver intelligent, context-aware applications. Proven track record of driving performance optimization, reliability, and end-to-end ownership in enterprise-scale systems.

TECHNICAL SKILLS

Languages & Frameworks: Java 8/11/17/21, Spring Boot, Spring Batch, Spring Security, React.js, LangChain4j

Cloud & Infrastructure: AWS (Lambda, EC2, ECS, EKS, S3, Fargate, Route53, CloudFront), PCF, Open-Shift, Docker, Kubernetes

Databases: PostgreSQL, MongoDB, Oracle

Architecture & Design: Microservices, REST API, Serverless, Event-Driven Architecture, System Design (HLD/LLD), OAuth 2.0

DevOps & Tools: Jenkins, GitHub Actions, Harness, Kafka, JUnit, BDD/TDD, Reactive Programming

Specializations: Generative AI Integration (OpenAI APIs), IoT Platforms, Healthcare Systems, Financial Services

GENERATIVE EXPERIENCE

- Designed and implemented **Retrieval-Augmented Generation (RAG)** pipelines using embeddings, vector databases, and LLMs for intelligent query answering.
- Integrated **OpenAI APIs** with Java microservices using **LangChain4j** to enable context-aware responses and reduce manual
- Implemented **prompt engineering strategies** to improve accuracy, reduce hallucination, and optimize LLM
- Developed AI-driven features such as **chatbots, summarization, and contextual search** for enterprise applications

PROFESSIONAL EXPERIENCE

Technical Lead — Cognizant (CVS Healthcare)

Aug 2024 – Present

- Architected and developed healthcare applications using Java 21, Spring Boot, and reactive programming, processing patient data for 10M+ users with 99.9% uptime
- Pioneered Generative AI integration in Java microservices leveraging OpenAI APIs and LangChain4j, delivering intelligent healthcare insights and reducing manual processing by 40%
- Designed serverless solutions using AWS Lambda and deployed on PCF, achieving 60% cost reduction in infrastructure spend
- Established robust CI/CD pipelines using Jenkins, GitHub Actions, and Harness, reducing deployment time by 70%
- Mentored team of 5 engineers in cloud-native development, microservices patterns, and AI/ML integration best practices
- Optimized PostgreSQL queries and implemented efficient data models, improving response times by 45%

Technical Lead — Siemens (Industrial IoT)

Jan 2023 – Aug 2024

- Led team of 6 engineers in developing enterprise-grade Industrial IoT platform serving 500+ industrial clients globally
- Architected scalable microservices using Java 8, Spring Boot, processing 1M+ IoT events daily with sub-100ms latency
- Implemented serverless architecture with AWS Lambda and PCF, reducing operational costs by 50% while improving scalability
- Designed and optimized PostgreSQL database schemas for global data distribution, ensuring ACID compliance across regions
- Automated deployment workflows with Jenkins CI/CD, achieving 95% deployment success rate .

Senior Associate Consultant — Infosys (Citizens Bank)*Mar 2020 – Jan 2023*

- Led development of critical banking applications using Java 8, Spring Boot microservices, handling 2M+ daily transactions
- Deployed and managed applications on OpenShift/Red Hat servers, ensuring 99.95% availability for mission-critical services
- Architected MongoDB data models for high-throughput banking operations, optimizing query performance by 55%
- Implemented comprehensive CI/CD pipelines with Jenkins, achieving 80%+ test coverage and automated security scanning
- Conducted security assessments using Fortify Scan, resolving 100% critical vulnerabilities and ensuring regulatory compliance
- Mentored junior developers in microservices architecture, containerization, and DevOps practices

Senior Software Engineer — Mindtree (SITA Airlines, JW Marriott)*May 2019 – Feb 2020*

- Delivered customized enterprise solutions for airline and hospitality clients using Java 8, Spring Boot, and microservices
- Deployed applications on IBM WebSphere and OpenShift, managing multi-environment deployments across cloud infrastructure
- Optimized SQL database performance, reducing query execution time by 40% through indexing and query optimization
- Implemented automated CI/CD pipelines, accelerating release cycles by 50%

Software Engineer — Mphasis (Charles Schwab)*Nov 2016 – Apr 2019*

- Developed financial services applications using Java 7, Spring frameworks, and Maven on WebSphere Application Server
- Managed Oracle databases using Toad, ensuring data integrity and optimizing complex queries for trading systems
- Established JUnit testing framework achieving 80%+ code coverage and implemented Veracode security scanning
- Collaborated with cross-functional teams to deliver features for 500K+ active users

EDUCATION**Master of Computer Applications (MCA)** — Techno India, Kolkata*2016***Bachelor of Computer Applications (BCA)** — Techno India College of Technology, Kolkata*2016*